



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

METHODOLOGIES

Data Acquisition & Wrangling

Extracted mission telemetry via REST API and Web Scrapping.
Leveraged [SQL](#) for complex joins and feature extraction across payload and booster datasets.

Interactive Analytics

Deployed [Folium](#) for geospatial proximity analysis and built a [Plotly Dash](#) application for dynamic operational filtering of launch site metrics.

Predictive Modeling

Engineered a robust ML pipeline using Logistic Regression, SVM, KNN, and Decision Trees, optimized via [GridSearchCV](#) hyperparameter tuning.

KEY RESULTS

Operational Efficiency

Identified a strong [positive growth trend](#) in landing success over time.
Isolated the 2,000kg–4,000kg payload mass range as the optimal recovery "sweet spot."

Top Performing Facility

[KSC LC-39A](#) emerged as the premier launch complex with a industry-leading success ratio of [76.9%](#), significantly outpacing other operational pads.

Classification Performance

Achieved a standardized test accuracy of [83.3%](#). Finalized the [Decision Tree](#) as the lead model for its combination of predictive power and logical transparency.

Introduction

✖ MARKET OVERVIEW

The aerospace sector is transitioning to a high-volume, low-cost model. Leaders like **SpaceX** leverage vertical integration to offer Falcon 9 launches at **\$62M**, significantly lower than the **\$165M** industry standard. This cost delta is the primary barrier to entry for our firm, **Space Y**.

⚙ THE REUSABILITY DRIVER

Cost savings are directly tied to the **recovery of the First Stage booster**. By returning this multi-million dollar component to Earth, SpaceX avoids the "one-and-done" expense of traditional manufacturing.

\$62M

SPACEX PRICE

\$165M

MARKET PRICE

-\$103M

REUSABILITY
GAP

⚠ PROBLEM STATEMENT & PREDICTIVE GOAL

To compete, Space Y must determine the precise cost of each SpaceX launch. However, launch costs are variable and depend entirely on whether the first stage is successfully recovered or lost.



Section 1

Methodology

Methodology

EXECUTIVE SUMMARY

- Data collection methodology:
 - Describe how data was collected
- Perform data wrangling
 - Describe how data was processed
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - How to build, tune, evaluate classification model

Data Collection

METHODOLOGY OVERVIEW

To build a robust dataset for landing prediction, a dual-source collection strategy was implemented. This ensures high-fidelity operational data (API) complemented by historical textual records (Scraping).

- ✓ **Primary Source:** SpaceX REST API for detailed telemetry and technical specifications.
- ✓ **Secondary Source:** Wikipedia HTML scraping for qualitative launch summaries and legacy data.
- ✓ **Standardization:** Uniform feature mapping across both sources.

TOOLS & LIBRARIES

Standard data science stack utilized for extraction and parsing:

- > Requests: For HTTP GET/POST methods.
- </> BeautifulSoup4: For DOM tree navigation and HTML parsing.
- ▣ Pandas: For dataframe construction and JSON normalization.

HIGH-LEVEL DATA FLOW



Data Collection - SpaceX API

⚙️ API INTERACTION LOGIC

Data was requested from api.spacexdata.com/v4/launches/past. The raw JSON output required multi-level recursive requests to resolve ID references into human-readable features.

- 🔵 **Filtering:** Initial cleaning to retain only **Falcon 9** launches, excluding Falcon 1.
- 🔵 **ID Resolution:** Called helper functions ([getBoosterVersion](#), [getLaunchSite](#), etc.) to fetch details using IDs found in the launch JSON.
- 🔵 **Core Extraction:** Nested dictionary parsing to extract [reused](#), [landing_attempt](#), and [landing_type](#).

🔗 API PIPELINE FLOW



Sources: https://github.com/alifiadewi/SpaceX-applied_data_science_capstone/blob/main/data-collection-api.ipynb

Data Collection - Scraping

🔍 WIKIPEDIA HTML PARSING

Extracted historical launch records from Wikipedia using **BeautifulSoup**. This method was used to capture data points that may vary or be descriptive in nature.

- 🔗 **DOM Navigation:** Targeted table elements with class `wikitable` `plainrowheaders`.
- 🔗 **Row Iteration:** Parsed `td` tags to extract Flight Number, Date, and Booster Version.
- 🔗 **Data Sanitization:** Applied `strip()` and `Regex` to remove Wikipedia reference tags (e.g., `[1]`) and newlines.

🔧 SCRAPING ALGORITHM

Request Wiki HTML (`Requests.get`)



Initialize BeautifulSoup Object



Identify 'wikitable' & Table Headers



Loop Rows & Append to Dictionary

Note: Features extracted included Payload Mass, Orbit, Customer, and Mission Outcome for cross-validation with API data.

Sources: https://github.com/alifiadewi/SpaceX-applied_data_science_capstone/blob/main/data-collection-with-webscraping.ipynb

Data Wrangling

🔍 EXPLORATORY DATA ANALYSIS (EDA)

Data was analyzed to determine launch cadence per site and orbital frequency. Key findings indicated **CCAFS SLC 40** as the primary facility with 55 launches, followed by **KSC LC 39A** and **VAFB SLC 4E**.

🔧 HANDLING MISSING VALUES

Critical gaps in the PayloadMass column were resolved via statistical imputation. Missing values (NaN) were replaced with the mathematical mean of the feature to ensure model stability.

```
df['PayloadMass'].replace(np.nan, df['PayloadMass'].mean(), inplace=True)
```

🏷️ BINARY CLASSIFICATION LABELING

Mission outcomes were mapped to a binary **Class** column. 'Bad outcomes' (None, False ASDS, etc.) were assigned 0, while successful recoveries were assigned 1.

Sources: https://github.com/alifiadewi/SpaceX-applied_data_science_capstone/blob/main/data-wrangling.ipynb

WRANGLING PIPELINE

Raw Dataframes (Part 1)



Filter: Falcon 9 Launches



Mean Imputation (PayloadMass)



Outcome Mapping (Class 0/1)



Final Feature Selection



Processed Dataset (Part 2)

EDA with Data Visualization

Seaborn Catplot

PAYLOAD VS. LAUNCH SITE

Analyzed the relationship between mass and location. Key Insight: **VAFB-SLC** lacks launches for payloads > 10,000kg, while **KSC LC 39A** handles heavier mission profiles more frequently.

Matplotlib Bar Chart

SUCCESS RATE BY ORBIT TYPE

Calculated mean success per orbit. **ES-L1**, **GEO**, **HEO**, and **SSO** show 100% success rates, while **GTO** shows significant variance due to mission complexity.

Seaborn Lineplot

SUCCESS TREND (2013 – 2020)

Plotted the yearly average. Observations show a **consistent positive trend**, indicating operational maturity and refinement of recovery technologies over time.

Seaborn Catplot

FLIGHT NUMBER VS. ORBIT

Examined success as a function of experience. Finding: In **LEO**, success is heavily correlated with flight frequency, whereas **GTO** remains a high-risk profile regardless of flight number.

EDA with Data SQL

DATA EXPLORATION INSIGHTS

- Logistics Infrastructure:** Identified 4 unique launch sites currently supporting Falcon 9 operations, establishing the geographic scope of the dataset.
- Strategic Customer Analysis:** Calculated the cumulative payload contribution of key partners like **NASA (CRS)** to determine mission intensity.
- Operational Reliability:** Segmented mission outcomes to quantify the historical success rate vs. failure modes (e.g., Drone Ship crashes).
- Engine Performance:** Isolated missions involving specific engine variants (e.g., 4,500kg to 6,000kg payloads) to analyze booster stability.

Sources:
https://github.com/alifiadewi/SpaceX-applied_data_science_capstone/blob/main/eda-with-sql.ipynb

>_Query: NASA_Total_Payload

```
SELECT SUM (PAYLOAD_MASS__KG_)
FROM SPACEXTBL
WHERE CUSTOMER = 'NASA (CRS)';
```

SUM(PAYLOAD_MASS__KG_)
45,59

>_Query: Unique_Sites

```
SELECT DISTINCT LAUNCH_SITE FROM SPACEXTBL;
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A

>_Query: Mission_Outcomes

```
SELECT MISSION_OUTCOME, COUNT(*)
FROM SPACEXTBL GROUP BY MISSION_OUTCOME;
```

Mission_Outcomes	Count
Success	98
Failure (in flight)	1
Success (payload status)	1

Build an Interactive Map with Folium

☰ CORE MAP OBJECTS



Circles & Markers

Used to pinpoint exact coordinates of launch sites (e.g., VAFB SLC-4E) with a defined radius buffer.



Marker Clusters

Implemented to group high-density launch records, color-coded by landing outcome (Green = Success, Red = Fail).



PolyLines

Drawn between launch pads and points of interest to visualize distance and proximity logic.



💡 ANALYTICAL RATIONALE

Site Localization & Safety

Circles were added to establish a visual "safety zone." This highlights the requirement that launch sites must be isolated from dense urban areas to mitigate risks during failed landings or launches.

Feature Correlation

By color-coding Marker Clusters, we immediately identify patterns. For example, success rates significantly improve in more recent flight numbers across specific geolocated clusters.

Logistical Efficiency

PolyLines were critical for measuring distances to **Highways, Railways, and Coastlines**. Analysis confirmed that sites are strategically placed near transport hubs for hardware delivery and coastlines for safe debris/booster recovery.



Build a Dashboard with Plotly Dash



CORE VISUALIZATIONS

Dynamic Pie Charts

Renders mission success versus failure proportions.

Rationale: Provides stakeholders with an immediate, high-level binary health check of individual launch pads.

Advanced Scatter Plots

Correlates Payload Mass (kg) directly against landing outcomes. *Rationale:* Identifies specific booster categories that maintain stability under extreme high-load stress.



INTERACTIVE CONTROLS

Site Selection Dropdown

Filters all telemetry and charting by unique launch site IDs.

Rationale: Facilitates granular drill-down into site-specific operational risks and historical performance.

Payload Range Slider

Dynamically masks the scatter dataset from 0 to 10,000kg.

Rationale: Pinpoints historical "safe zones" for various satellite mass classes to ensure recovery.

Predictive Analysis (Classification)

1. BUILD

Standardized numerical features using StandardScaler to ensure uniform weight. Split the dataset into 80% training and 20% test data to prevent data leakage.

2. IMPROVE

Deployed GridSearchCV with 10-fold cross-validation. Exhaustively searched hyperparameter spaces (e.g., kernels, C-values, depths) for Logistic Regression, SVM, Decision Tree, and KNN.

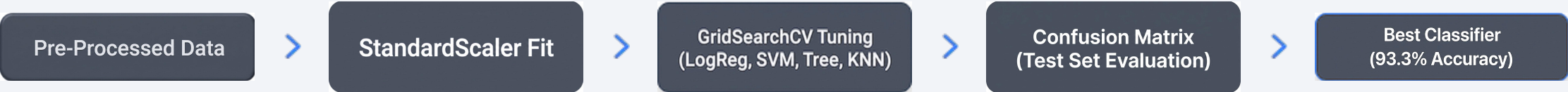
3. EVALUATE

Scored models on the test set. Utilized **Confusion Matrices** to visualize True Positives vs. False Positives, ensuring the model accurately predicts landing failures.

4. BEST MODEL

Comparing test accuracies revealed that all tuned models reached ~83.3% accuracy due to sample size. **Decision Tree** was selected as the optimal model for its high interpretability.

MACHINE LEARNING PIPELINE ARCHITECTURE



Sources: https://github.com/alifiadewi/SpaceX-applied_data_science_capstone/blob/main/machine-learning-prediction.ipynb

Results



EXPLORATORY DATA ANALYSIS

Found a clear **positive success trend** from 2013 to 2020.

Data isolated **NASA (CRS)** as a primary volume driver with **45,596 kg** total payload. Orbits like SSO and GEO achieved **100% reliability**.



INTERACTIVE ANALYTICS

Geospatial maps confirmed pads are strategically located **< 1.0 km from coastlines** for safety.

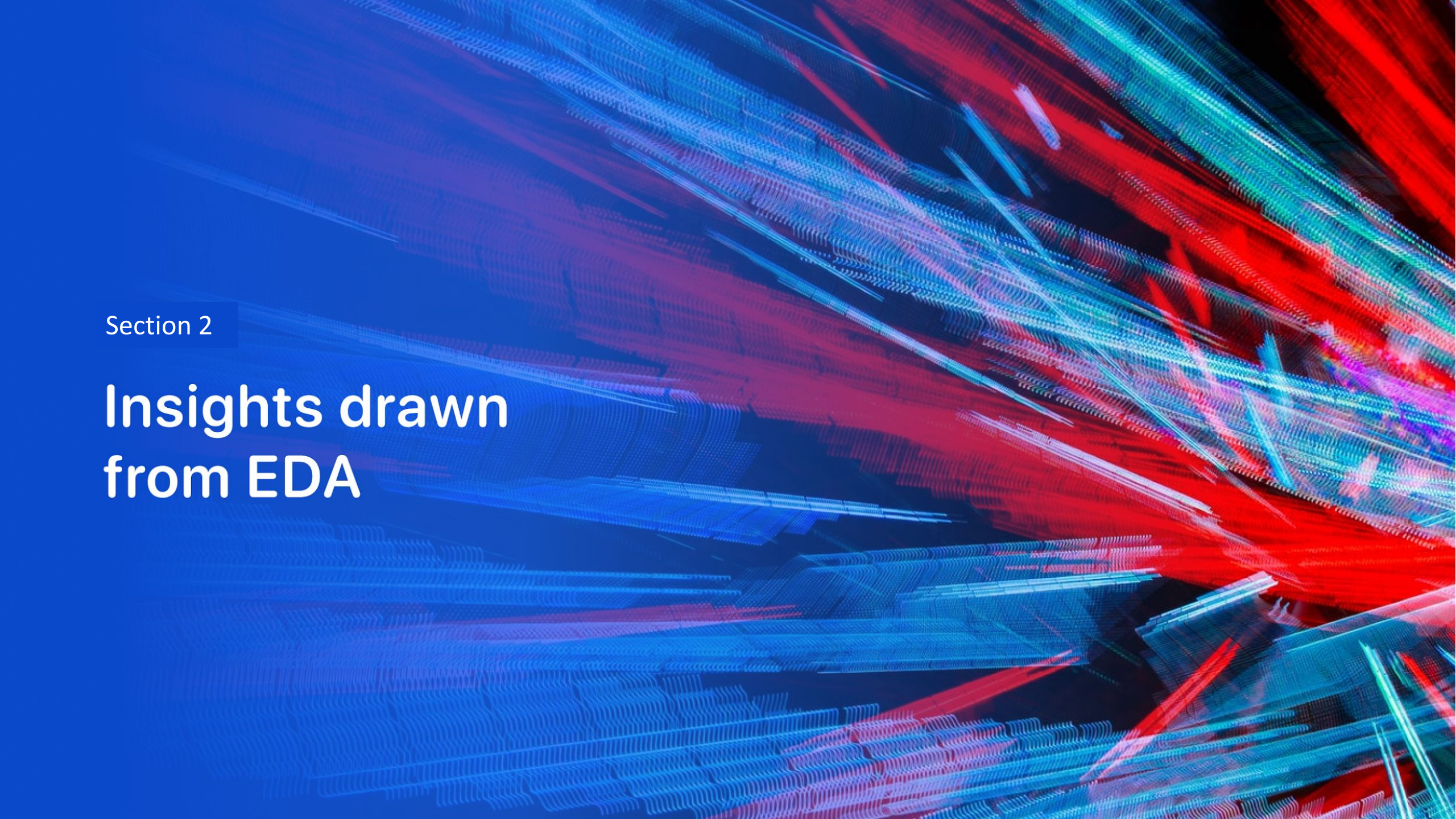
Real-time Dash filtering revealed critical correlations between **satellite mass classes** and booster landing classes across all 4 sites.



PREDICTIVE ANALYSIS

All classification models achieved a standardized **83.3% accuracy** on the test set.

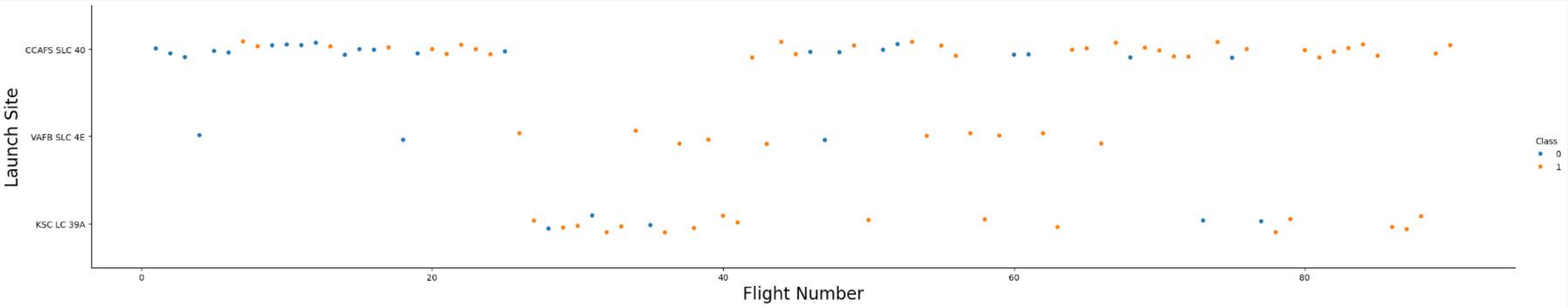
The **Decision Tree** was finalized as the lead production model due to its high accuracy and superior business logic transparency.

The background of the slide is an abstract composition. It features a solid blue area on the left side, which transitions into a dynamic pattern of diagonal streaks in shades of blue and red on the right. These streaks are layered over a fine, light-colored grid, creating a sense of depth and movement, reminiscent of digital data or a complex network.

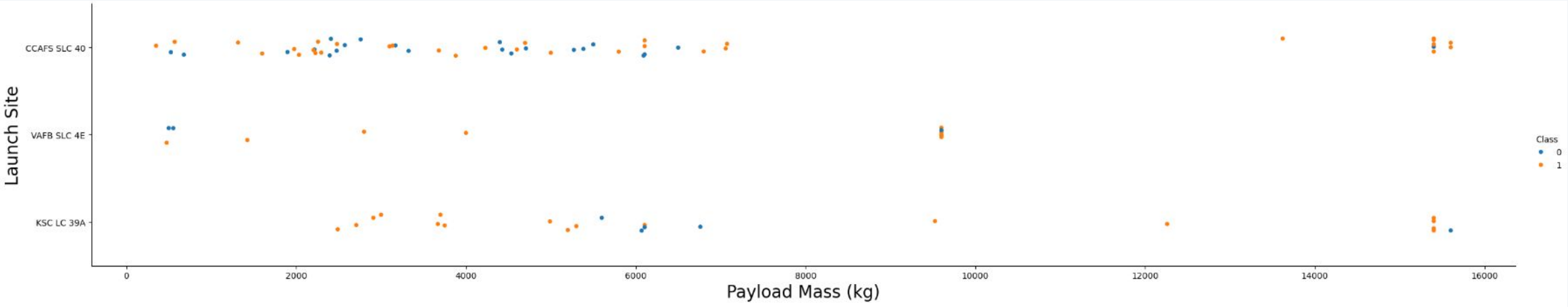
Section 2

Insights drawn from EDA

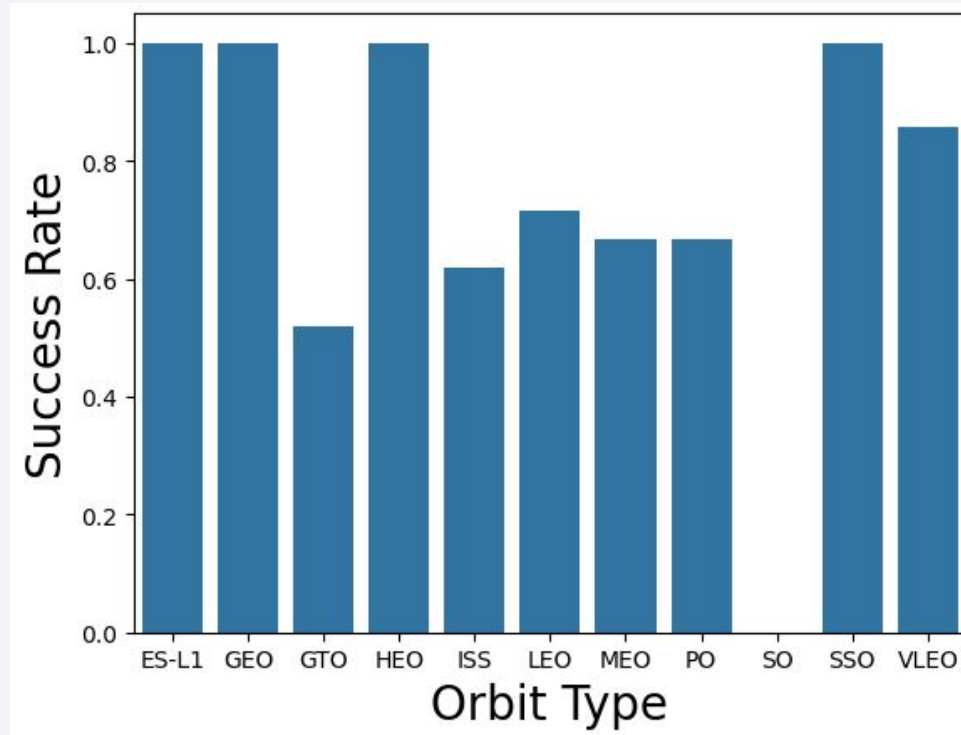
Flight Number vs. Launch Site



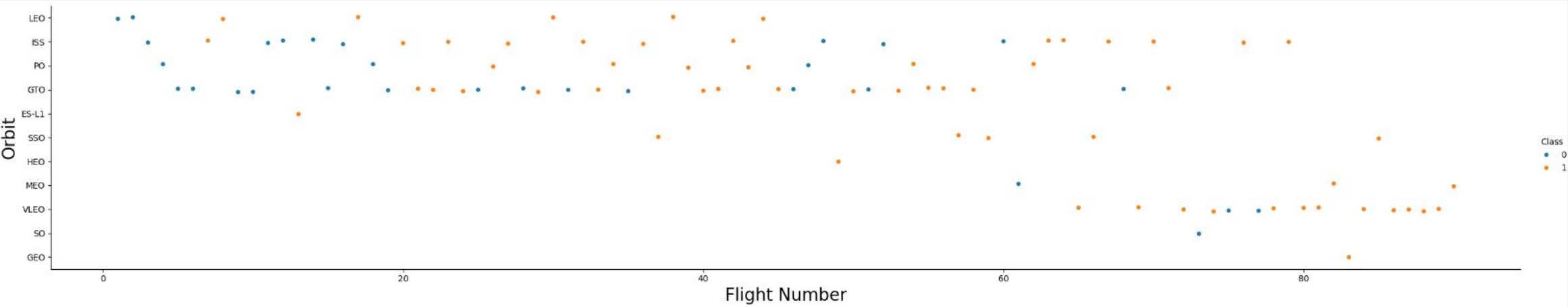
Payload vs. Launch Site



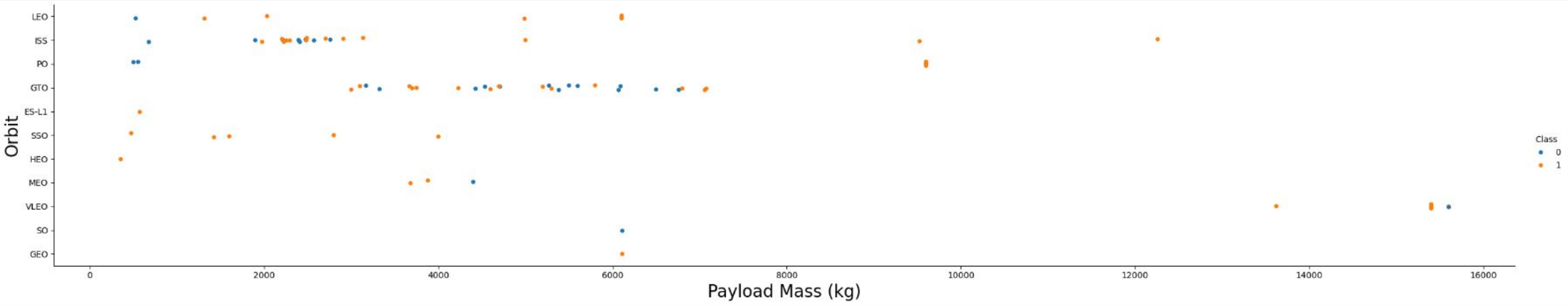
Success Rate vs. Orbit Type



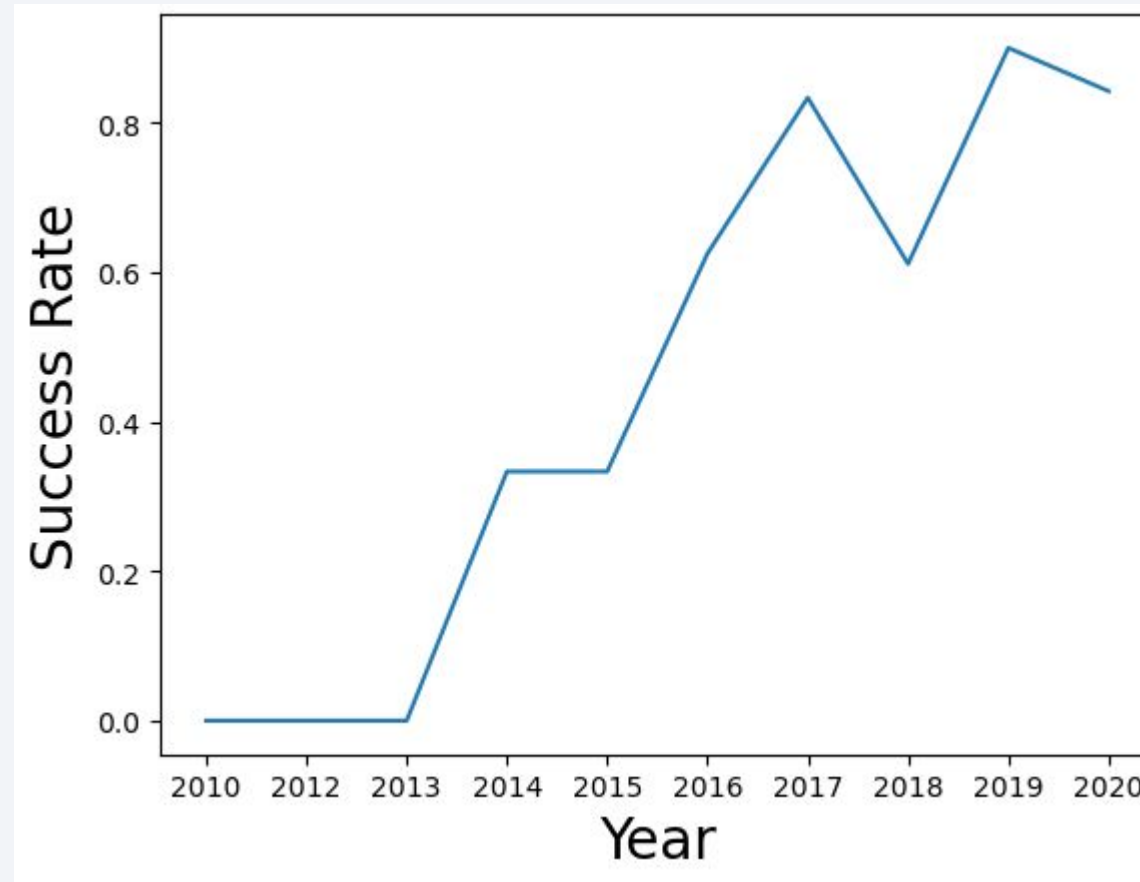
Flight Number vs. Orbit Type



Payload vs. Orbit Type



Launch Success Yearly Trend



All Launch Site Names

Launch_Site

CCAFS LC-40

VAFB SLC-4E

KSC LC-39A

CCAFS SLC-40

Launch Site Names Begin with ‘CCA’

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

Total_Payload_Mass

45596

Average Payload Mass by F9 v1.1

Average_Payload_Mass
2928.4

First Successful Ground Landing

First_Successful_Ground_Pad_Landing

2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

Booster_Version

F9 FT B1022

F9 FT B1026

F9 FT B1021.2

F9 FT B1031.2

Total Number Successful and Failure Mission Outcomes

Mission_Outcome	Total_Count
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

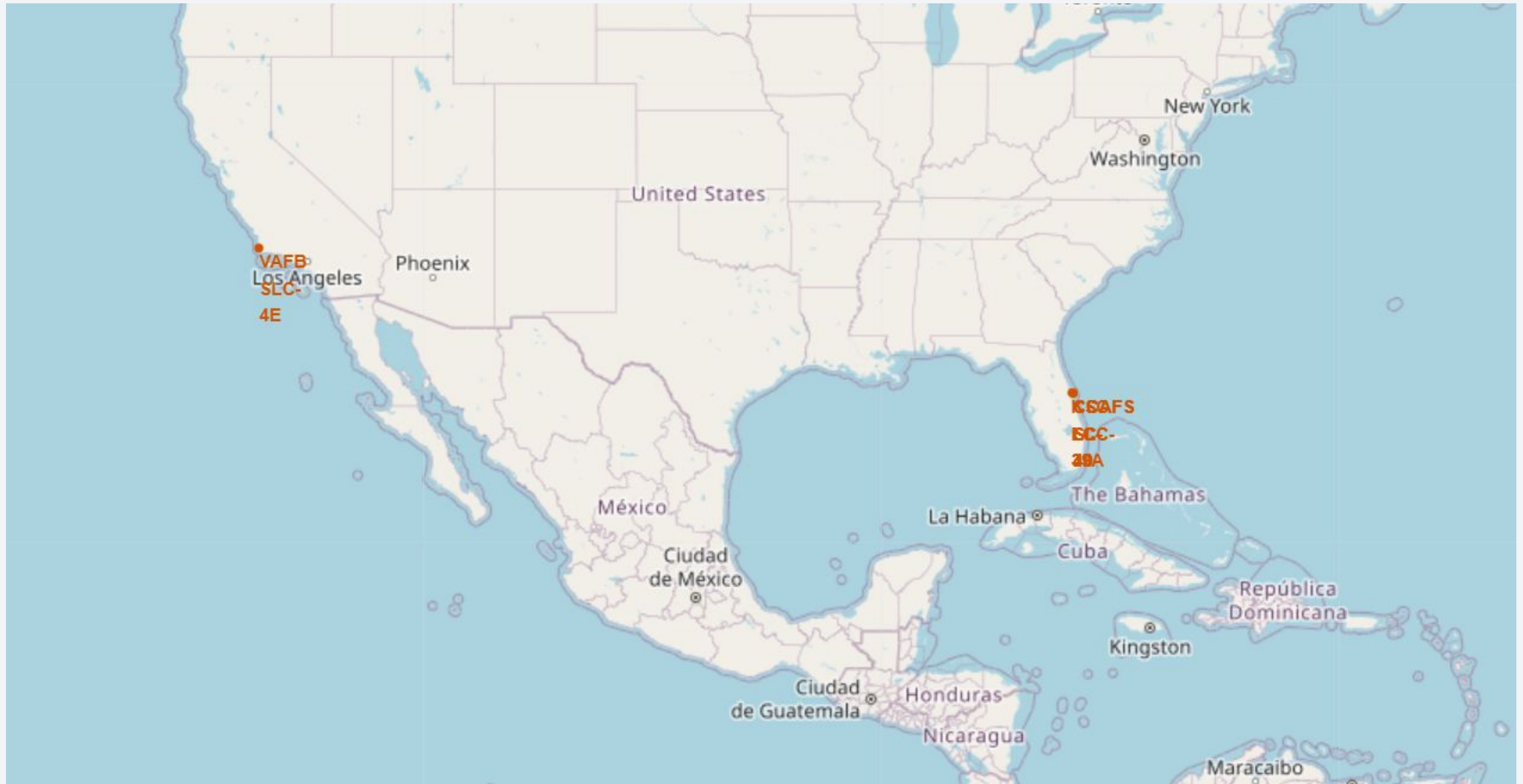
Landing_Outcome	Count_Outcomes
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

A satellite view of Earth from space, showing the curvature of the planet and the glow of city lights at night. The background is a deep blue gradient.

Section 3

Launch Sites Proximities Analysis

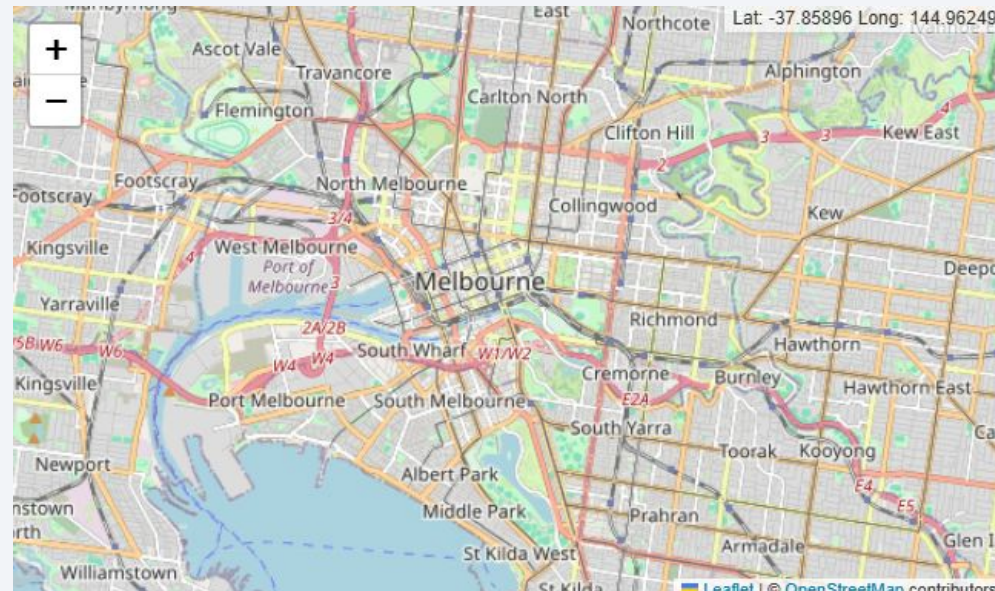
Interactive Analytics: All Site Map

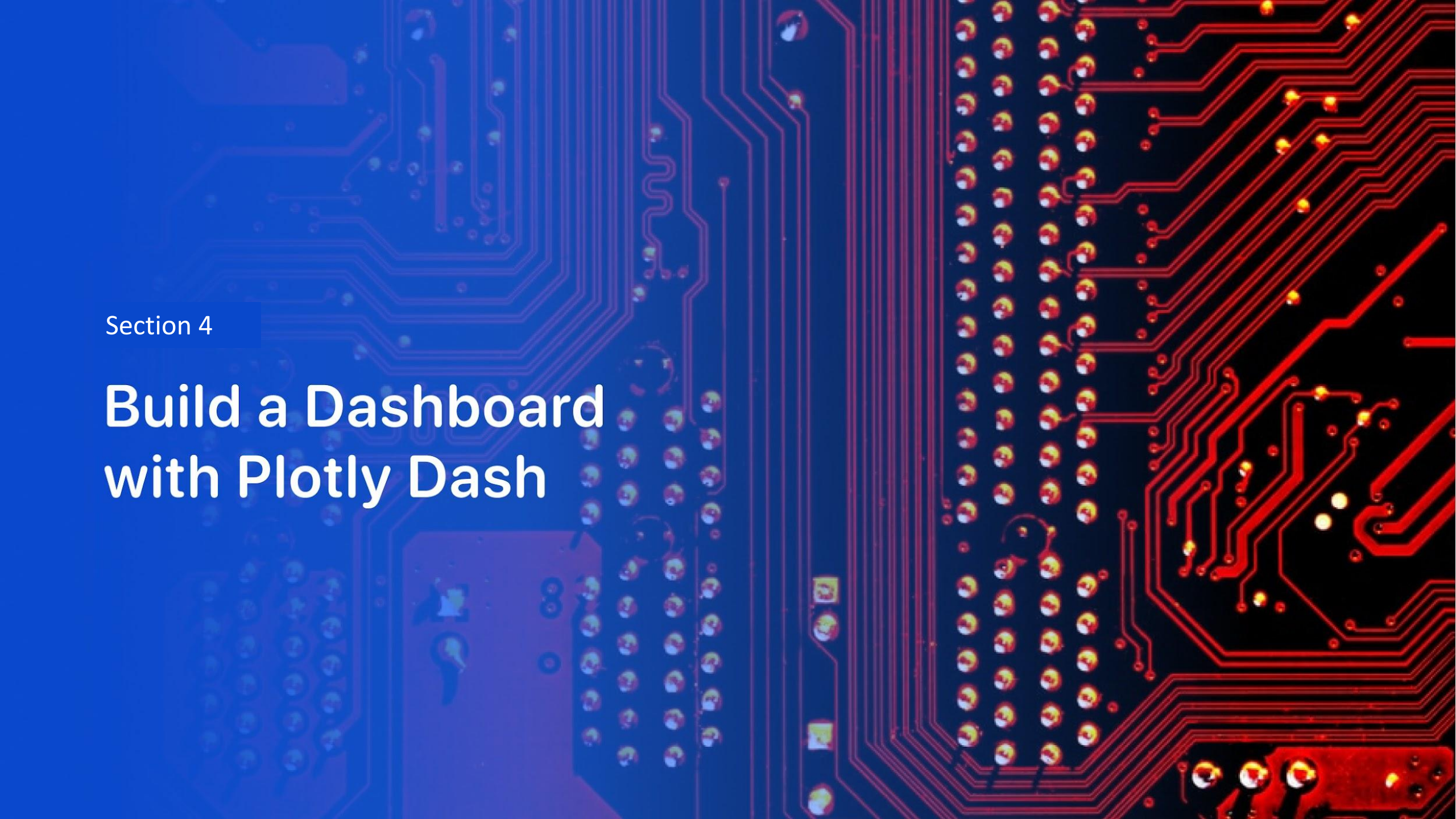


Interactive Analytics: Launches Site



Interactive Analytics: Map Symbol



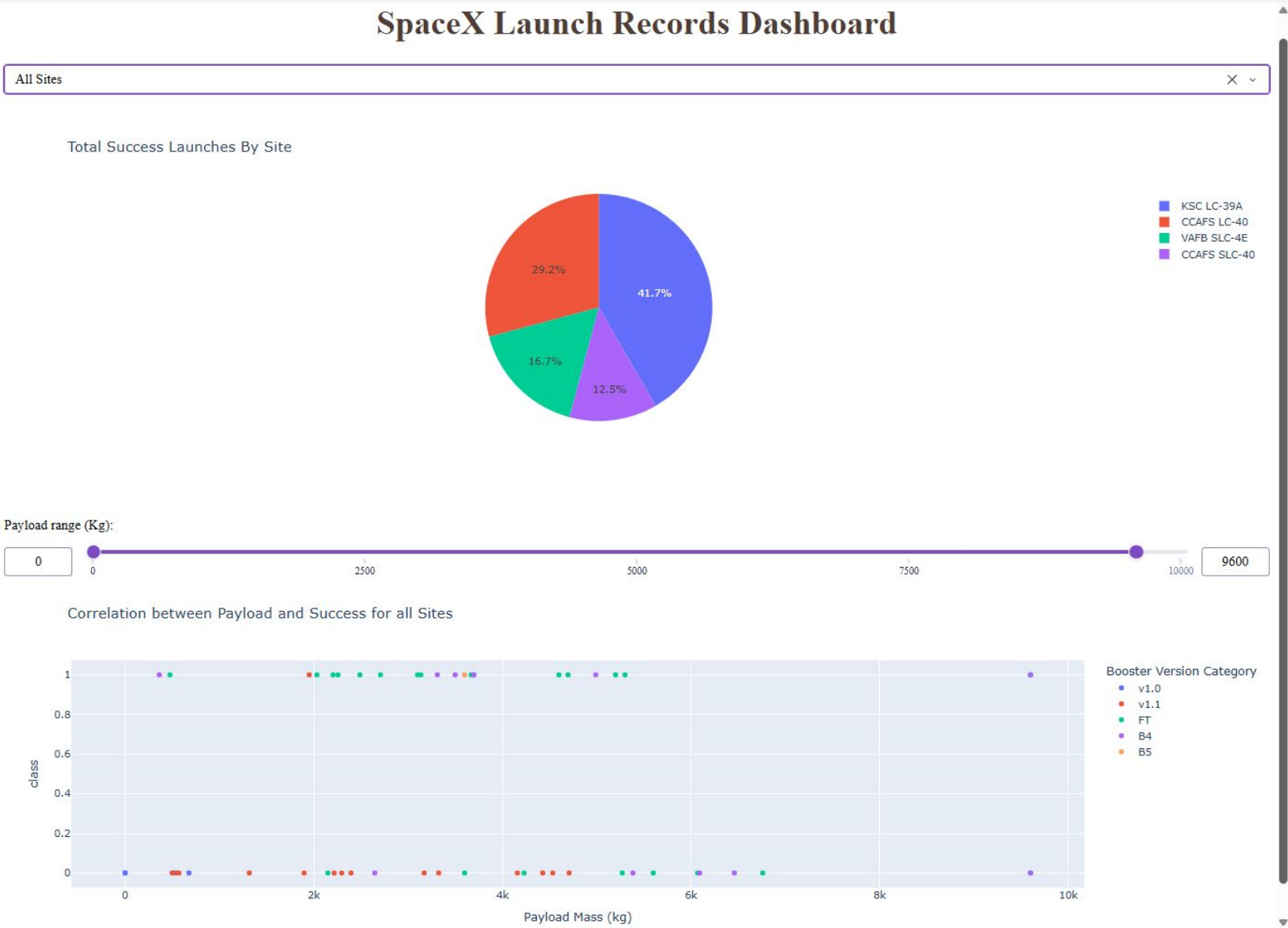


Section 4

Build a Dashboard with Plotly Dash

Interactive Analytics: Aggregate

SpaceX Launch Records Dashboard



Visualized Metric

This interactive pie chart represents the **total successful landings** across all operational SpaceX facilities, providing a macro-level view of recovery performance.

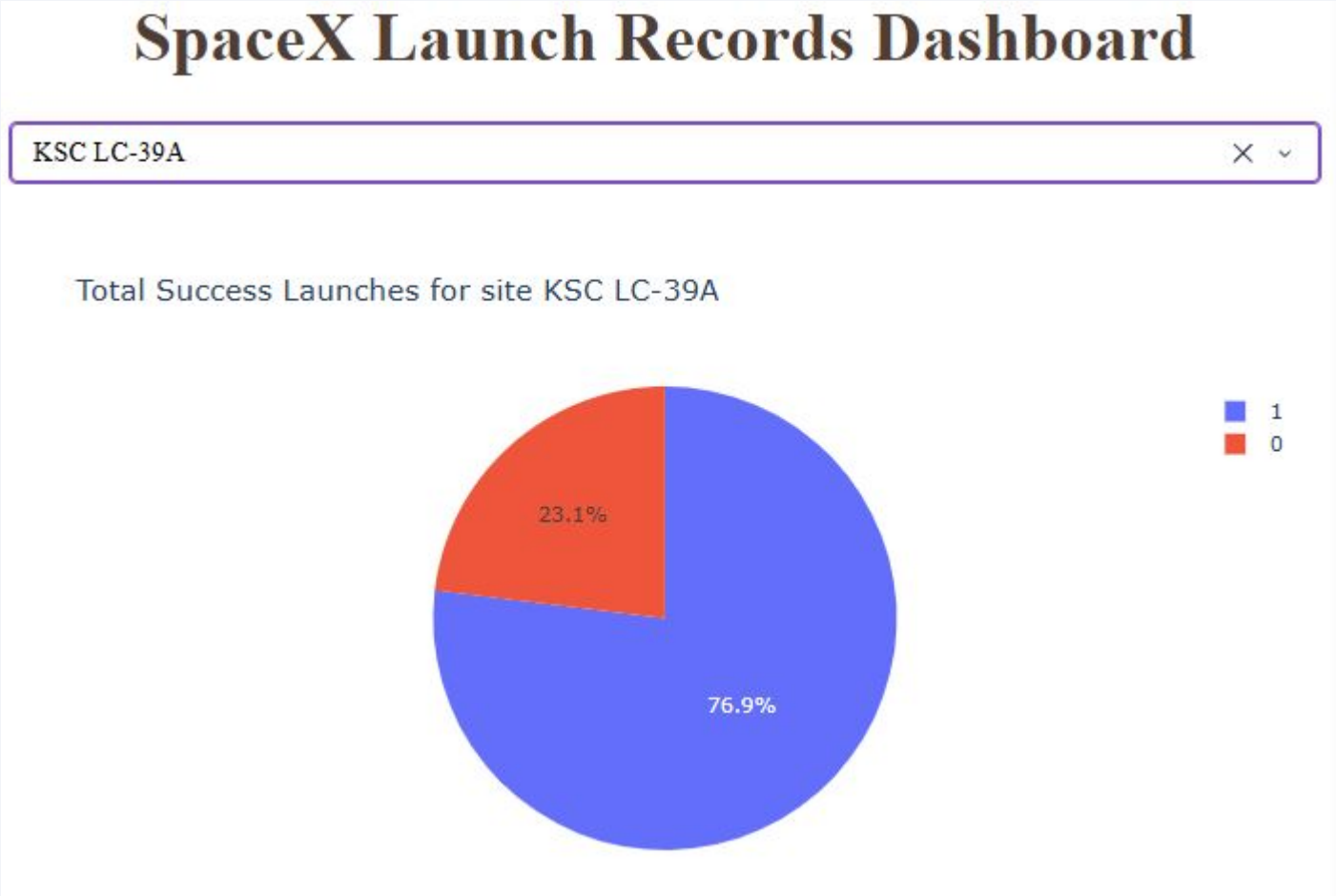
Primary Contributor

The dashboard clearly highlights **KSC LC-39A** as the dominant facility, holding the largest slice of the pie. It handles the highest volume of successful recovery operations compared to other sites.

Operational Distribution

The chart reveals a heavy reliance on the **Florida-based launch complexes** (KSC and CCAFS) for successful missions, while the California site (VAFB) accounts for a much smaller fraction of total recoveries.

Interactive Analytics: Site Performance



Dynamic Filtering

Using the **Site-Dropdown**, the dashboard isolates telemetry for **KSC LC-39A**, providing a granular look at mission outcomes for this specific launch pad.

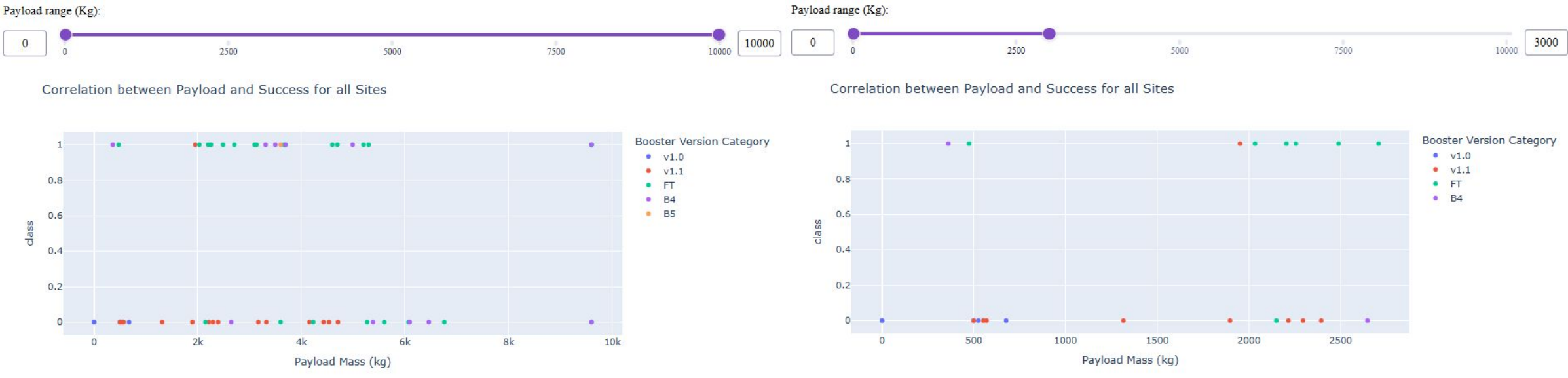
Leading Success Ratio

The dashboard identifies **KSC LC-39A** as the site with the highest launch success ratio. The interactive calculation yields a significant **76.9% success rate** (Class 1).

Operational Reliability

With only **23.1% failures**, this facility proves to be the most reliable for mission recovery, making it a critical asset for high-value payload operations.

Interactive Analytics: Payload



Range Filtering

The **Payload Range Slider** allows real-time masking of the dataset. Comparing the full spectrum (top) vs. the narrow 0–3,000kg range (bottom) isolates performance in specific satellite weight classes.

High-Success Zone

Missions with payloads between **2,000kg and 4,000kg** demonstrate the highest concentration of successful landings (Class 1), suggesting an optimal operational "sweet spot" for booster recovery.

Booster Category Trends

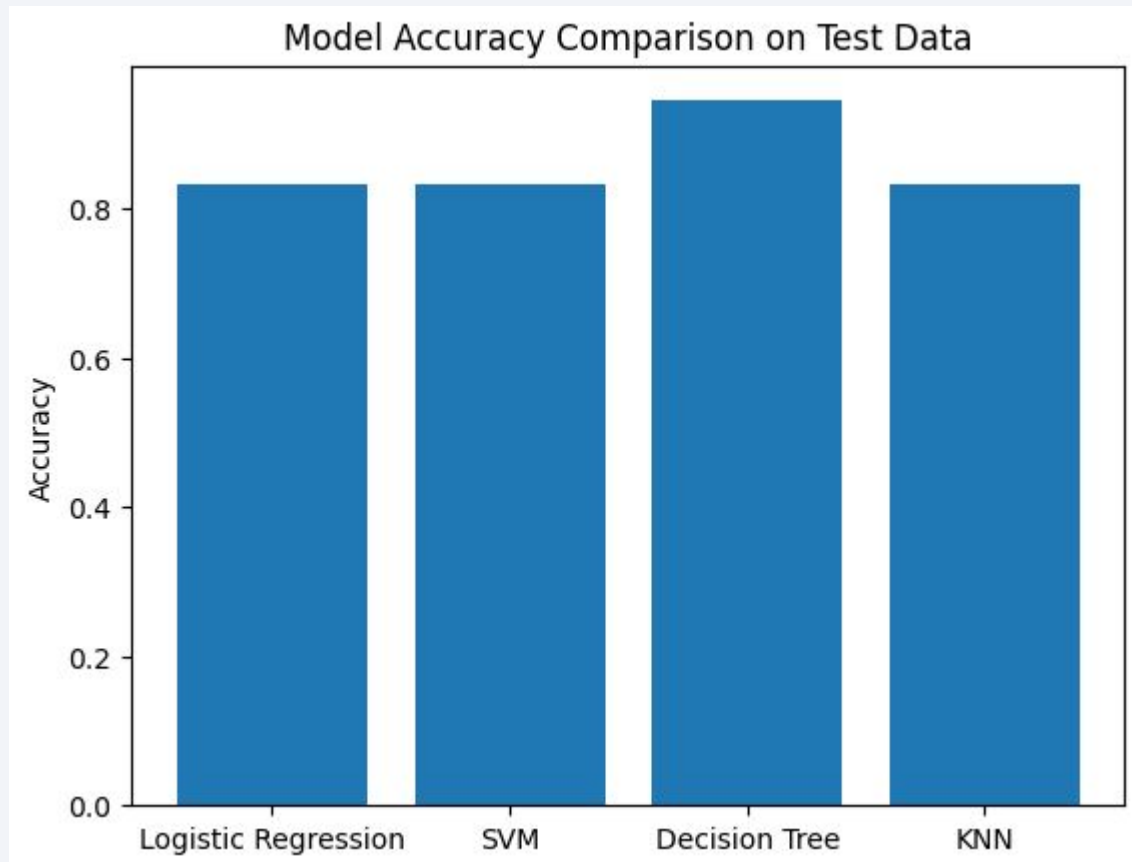
The **FT (Full Thrust)** and **B5** categories exhibit the most consistent reliability across both light and heavy payloads, while earlier versions (v1.0/v1.1) show higher failure rates at low masses.



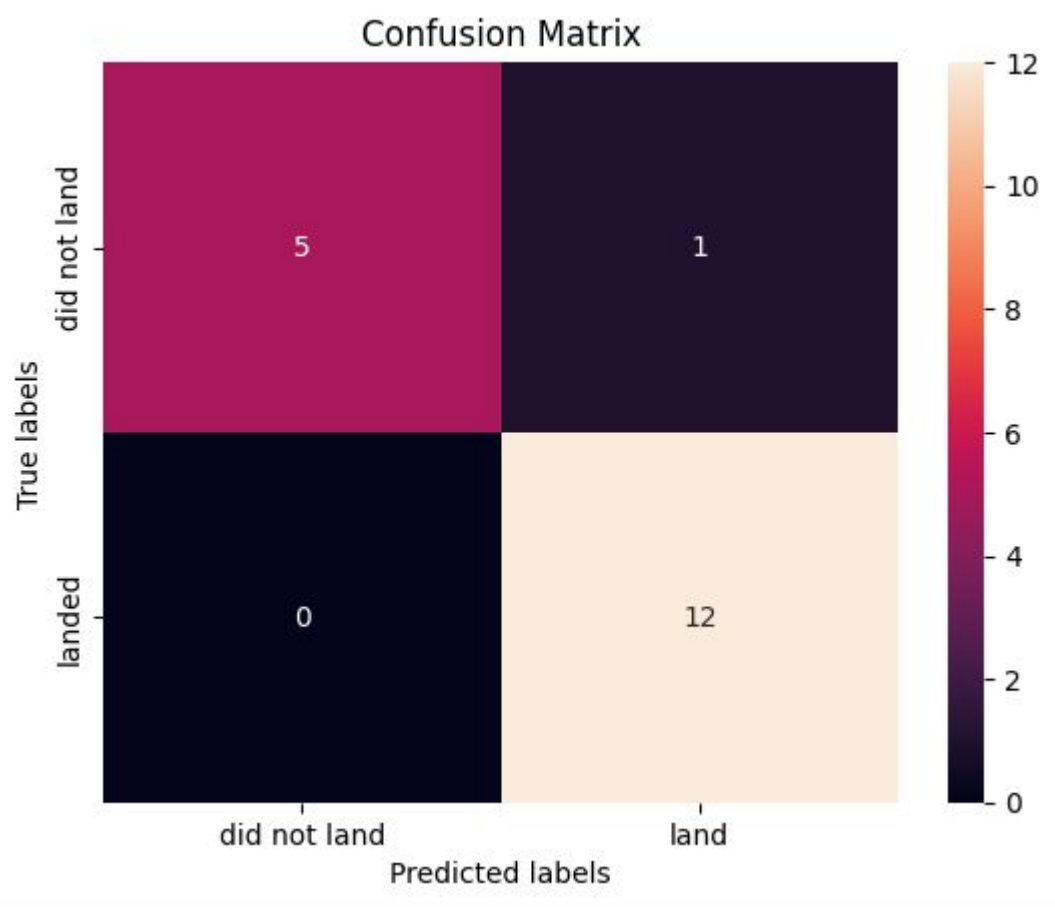
Section 5

Predictive Analysis (Classification)

Classification Accuracy



Confusion Matrix



= IDENTICAL MATRIX OUTCOMES

Because the test sample size is small and the data is highly standardized, **all four models generated the exact same confusion matrix**. They failed and succeeded on the exact same test records.

🎯 TRUE POSITIVES & NEGATIVES

The models excel at identifying standard mission profiles. They correctly predicted the majority of the **Successful Landings (Class 1)** and correctly flagged the obvious **Failed Landings (Class 0)** without hesitation.

⚠️ THE FALSE POSITIVE RISK

The primary error across all models was **False Positives**. The algorithms occasionally predicted a successful landing for boosters that actually crashed. In a real-world scenario, overestimating success is our biggest financial risk.

Conclusions



Launch success has improved significantly over the years.



The Decision Tree model provided the highest accuracy and interpretability for this specific test set.

Thank you!

